

3 Internal and Entrusted Manufacturing Technologies

This section defines the manufacturing primitives that later enter route values. It does not construct those values, choose a route, or clear the qualified manufacturing-service market. All technology primitives are invariant to the MAH regime. The only direct institutional wedge remains $\tau_E(M)$.

3.1 Internal manufacturing

For a project with manufacturing requirement $m > 0$ and a developer with internal manufacturing capability $k_i > 0$, let

$$c_I(m, k_i) > 0, \quad c_{I,m}(m, k_i) > 0, \quad c_{I,k}(m, k_i) < 0 \quad \text{on the interior of the feasible domain.} \quad (1)$$

The function c_I is technological marginal manufacturing cost, measured in C/Y. A more demanding project raises this cost, while greater internal capability lowers it on the feasible domain.

Internal production also requires a project-level production-readiness/setup cost $F_I(m, k_i)$, measured in C. Let $\underline{k}(m) > 0$ be the minimum capability needed for an internally produced project with requirement m . The feasibility convention is

$$\begin{aligned} 0 \leq F_I(m, k_i) < +\infty, & \quad \text{if } k_i \geq \underline{k}(m), \\ F_I(m, k_i) = +\infty & \quad \text{if } k_i < \underline{k}(m). \end{aligned} \quad (2)$$

On the interior of the feasible domain, assume smoothness and $F_{I,m} > 0$ and $F_{I,k} < 0$. Derivatives are not taken across the feasibility boundary. Thus low internal capability makes route I unavailable through an infinite setup cost. No separate binary internal-feasibility primitive is retained.

3.2 Qualified external production

Under entrusted manufacturing, a qualified external producer uses the technological marginal-cost kernel

$$c_E(m) > 0, \quad (3)$$

also measured in C/Y. This kernel does not depend on the developer's internal capability k_i : manufacturing is performed by the qualified external producer.

An entrusted project requires

$$b(m) > 0, \quad b'(m) > 0, \quad (4)$$

units of qualified manufacturing-service capacity. The object $b(m)$ is a physical capacity requirement in B per project. It is not a route share, a supply curve, or an exogenous measure of scarcity relief.

The real technology-transfer, validation, and production-readiness cost is

$$F_E(m) \geq 0, \quad (5)$$

measured in C per project. In addition, the developer bears a residual holder-side responsibility and coordination burden

$$\mu_E \geq 0. \quad (6)$$

The burden μ_E is not eliminated by MAH.

The accounting roles are distinct. The technological marginal cost $c_E(m)$ is the input to the operating-value kernel derived above. The separate monetary capacity payment $p_m b(m)$, the readiness cost $F_E(m)$, the holder burden μ_E , and the institutional barrier $\tau_E(M)$ are not embedded in $c_E(m)$. Each can therefore enter a later route value at most once.

3.3 Organizational and institutional distinction

Routes I and E are technologically and organizationally distinct:

- under I , the developer remains the holder and manufactures using its own capability k_i ;
- under E , the developer remains the authorization holder, retains holder responsibility, and a qualified external producer manufactures;
- route E is not the transfer or out-licensing route T , under which the retained holder–producer organization is not used.

The only direct policy object is

$$M \in \{0, 1\}, \quad \tau_E(0) = +\infty, \quad \tau_E(1) = \bar{\tau}_E < +\infty. \quad (7)$$

The regime does not change c_I , F_I , $\underline{k}$, c_E , b , F_E , μ_E , a_i , k_i , q , m , $s(q)$, or $s_g(q)$. In particular, this technology block does not posit a fall in p_m^* or an outward shift of CMO supply. Price and capacity are equilibrium objects determined in the qualified manufacturing-service market below.

3.4 Boundary checks

If $k_i < \underline{k}(m)$, internal production is infeasible because $F_I = +\infty$; this is the required low-capability boundary. Increasing k_i within the feasible domain lowers both internal marginal and setup costs, while increasing m raises both. The entrusted technology does not use k_i , but a larger m strictly raises its qualified-capacity requirement $b(m)$. None of these statements is a route-choice result: route values, outside options, and deterministic sorting are defined in the next section.